VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
(ELECTRICAL AND ELECTRONICS ENGINEERING)

VII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22PC1EE401	Switchgear and Protection	3	0	0	3	3
22PC1EC421	Principles of Digital Signal Processing	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1EE401	Power System Operation and Control	3	0	0	3	3
22PE1EE402	Switched Mode Power Conversion					
22PE1EE403	Electric Vehicle					
22PE1EE422	Sensors and Actuators					
22PE1EC302	Real Time Operating Systems					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Green Energy Stream	3	0	0	3	3
	Renewable Energy System Stream					
	Electronics Engineering Stream					
	Internet of Things Stream					
	VLSI Design Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2EE401	Power Systems Laboratory	0	0	2	2	1
22PC2EC421	Principles of Digital Signal Processing Laboratory	0	0	2	2	1
22PW4EE401	Major Project Phase – I	0	0	6	6	3
Total		**15**	**0**	**10**	**25**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
ELECTRICAL AND ELECTRONICS ENGINEERING

VIII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1EE404	HVDC Transmission	3	0	0	3	3
22PE1EE405	Microgrid Operation and Control					
22PE1EE406	Integration and Control of Renewable Technologies					
22PE1EE407	Reliability Engineering Applications to Power Systems					
22PE1EC420	CPLD and FPGA Architecture					
	PROFESSIONAL ELECTIVE – VI					
22PE1EE408	Power Quality and FACTS	3	0	0	3	3
22PE1EE409	Energy Auditing and Conservation					
22PE1AE306	Fuel Cell Technology					
22PE1EE410	High Voltage Engineering					
22PE1EE411	Smart Grid					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4EE402	Major Project Phase – II	0	0	22	22	11
Total		**9**	**0**	**22**	**31**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1EE401) SWITCHGEAR AND PROTECTION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Electrical Machines – II & III, Power System – II

COURSE OBJECTIVES:

- To understand the need of protection of electric equipment and their protection schemes
- To introduce all kinds of circuit breakers and relays for protection of Generators, Transformers, and feeder bus bars from Over voltages and other electrical hazards
- To understand operations & characteristics of various static relays
- To explore digital protection and applications of PMUs in power system protection
- To explore and applications of AI in power system Protection

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the operation of various types of circuit breakers and relays
CO-2: Application of various protection schemes of various power system components like alternators, transformers, and busbars
CO-3: Appreciate the operation of static relays
CO-4: Comprehend PMU and its application in power system
CO-5: Apply AI to Power system

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	1	2	3	3	3	2	3	-	-	-	2	2	2	2
CO-2	2	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-3	1	2	3	3	3	2	3	-	-	-	2	2	3	3
CO-4	2	2	3	2	3	2	-	-	-	-	2	2	3	3
CO-5	3	2	2	2	3	2	2	-	-	-	2	2	3	3

UNIT-I:
Circuit Breakers: Elementary principles of arc interruption, Recovery, Restriking Voltage Max. RRRV and Numerical Problems. CB ratings, Description and Operation of Minimum Oil Circuit breakers, Air Blast Circuit Breakers, Vacuum and SF6 circuit breakers.

UNIT-II:
Protective Relays: Essential qualities of protective relaying, concept of Electromagnetic Relays, Universal torque equation, over current, Directional relays, travelling wave based line directional (TW32) and Differential (TW87) Relays.

Distance Relays: Impedance, Reactance and Mho Relays, Characteristics of Distance Relays and Comparison.

UNIT-III:

Equipment Protection Schemes: Differential Protection scheme of Generators, Negative Sequence Relay, Protection of Transformers: Percentage Differential Protection, Numerical Problem on Design of CT's Ratio, Buchholz relay Protection, Differential Over-Current protection of Bus bars. Three-zone distance relay protection using Impedance relays, Relay Coordination, Peterson Coil Grounding

UNIT-IV:

Static Relays: Basic construction of Static Relays, Advantages of static relays – Instantaneous over current relay, General equation for comparators, Phase comparator, Amplitude Comparator - Duality between amplitude and phase comparators.

UNIT-V:

Digital Protection: Introduction to Numerical protection, Digital protection Technology Overview-Digital Relay Architecture, Signal path in Numerical Relay, Digital Relaying Algorithm, Fault location; Introduction to PMU and its use, Application of Artificial Intelligence to Power System Protection (Digital Relaying).

TEXT BOOKS:

1. Switchgear and Protection, J. B. Gupta, S. K. Katari & Sons, 2013
2. Static Relays, T. S. Madhava Rao, 2nd Edition, TMH, 1989
3. Computer Relaying for Power Systems, Arun G. Phadke and James S. Thorp, 2nd Edition, Wiley

REFERENCES:

1. Power System Protection and Switch Gear, Badri Ram and D. N. Vishwakarma, McGraw-Hill Professional
2. Switchgear Protection and Power System, Sunil S. Rao, 13rd Edition, Khanna Book Publishing Co. (P.) Ltd.
3. Fundamentals of Power Systems, Y. G. Paitankar, S. R. Bhinde, 2nd Edition, PHI Publication
4. The Art and Science of Protective Relaying, C. Russell Mason, Wiley-Blackwell Publishers, 1966

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/108/105/108105167/
2. https://nptel.ac.in/courses/108/107/108107167/
3. https://nptel.ac.in/courses/117/107/117107148/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1EC421) PRINCIPLES OF DIGITAL SIGNAL PROCESSING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Signals and Systems, Network Analysis, Analog and Digital circuits

COURSE OBJECTIVES:
- To understand characteristics of discrete time signals and systems
- To analyze and process signals using various transform techniques
- To understand various factors involved in design of digital filters
- To understand the features of TMS24XX processors

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand and analyze continuous, discrete signals and systems
CO-2: Analyze the transformation techniques and their computation
CO-3: Design digital filters from analog filters
CO-4: Realization of digital filters and understand the architecture of DSP Processors

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	2	2	1	2	2	3	1	1	1	1	3	2
CO-2	3	3	2	2	2	1	2	3	1	1	1	1	3	2
CO-3	3	3	3	2	3	1	2	3	1	1	1	1	3	2
CO-4	3	3	3	2	3	2	3	3	2	1	1	1	3	2

UNIT-I:
Introduction to Signals and Systems: Classification of Continuous time Signals & Systems. Linear shift invariant systems, stability and causality-Introduction to digital signal processing, Sampling of Continuous signals- -Sampling Process-Sampling Theorem. Classification of discrete time signals and sequences- Discrete time systems

UNIT-II:
Fourier Analysis: Introduction to Discrete Fourier series, Discrete Fourier Transform: Properties of Discrete Fourier Transform, linear convolution and circular convolution of sequences using DFT, Computation of DFT, Relation between DFT and Z-Transform.
Fast Fourier Transform: Radix -2 decimation in time and decimation in frequency FFT algorithms, Inverse FFT

UNIT-III:
Z- Transform: Introduction to Z-transform, Properties of Z- Transform, Inverse ZTransform, Application of Z- Transforms for Linear constant coefficient difference equations, Realization of Digital filters, system function – stability criterion.

UNIT-IV:
IIR Filters: Analog filter approximations-Design of Butterworth Chebyshev filters, Design of IIR digital filter from analog filter using- impulse invariant and bilinear transformation techniques, design examples, realization of IIR filters-direct, canonic, cascade, and parallel forms.

UNIT-V:
FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR filters using – Rectangular and Hamming windows, comparison of FIR and IIR filters, realization of FIR filters-direct, cascade forms.
Digital Signal Processors: General DSP processor architecture, TMS320C67 processor architecture, features, addressing modes, and applications.

TEXT BOOKS:

1. Digital Signal Processing: Principles, Algorithms and Applications, John G. Proakis, D. G. Manolakis, 3rd Edition, PHI, 2007
2. Discrete Time Signal Processing, A. V. Oppenheim, R. W. Schaffer, PHI, 2009
3. TMS 320F 24xx Manuals

REFERENCES:

1. Digital Signal Processing – Fundamentals and Applications, Li Tan, Elsevier, 2008
2. Fundamentals of Digital Signal Processing using MATLAB, Robert J. Schilling, Sandra L. Harris, Thomson, 2007
3. Digital Signal Processing, S. Salivahanan, A. Vallavaraj, C. Gnanapriya, TMH, 2009
4. Discrete Systems and Digital Signal Processing with MATLAB, Taan S. El Ali, CRC Press, 2009
5. Digital Signal Processor: Architecture, Programming & Application, P. Venkata Ramani, M. Bhaskar, Tata McGraw-Hill, 2001

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EE401) POWER SYSTEM OPERATION AND CONTROL

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Electrical Machines, Power Systems and Control Systems

COURSE OBJECTIVES:
- To understand the formulation of Load-Flow problems applying different methods and carryout load-flow studies and compare
- To understand the importance of Economic Operation of Power Systems including losses
- To understand the importance of Load Frequency Control in the operation of power systems
- To understand the importance of reactive power and role of SCADA for stable operation of Power systems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the Optimal operation of Hydro and Thermal power plants
CO-2: Develop the transfer function models and block diagrams of single and two area systems
CO-3: Evaluate the performance of single and two area systems with and without Controllers
CO-4: Assess the behavior of Reactive power compensation devices

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	3	2	1	3	1	1	-	-	-	2	1	3	2
CO-2	2	3	2	1	3	1	1	-	-	-	2	1	3	2
CO-3	3	3	3	2	3	2	2	-	-	-	2	1	2	3
CO-4	3	2	2	1	2	-	1	-	-	-	2	1	3	2

UNIT-I:
Economic Operation of Power Systems: Optimal operation of Generators in Thermal Power Stations, - heat rate Curve – Cost Curve – Incremental fuel and Production costs, input-output characteristics, Optimum generation allocation with line losses neglected. Optimum generation allocation including the effect of transmission line losses – Loss Coefficients, General transmission line loss formula.

UNIT-II:
Unit Commitment and Hydrothermal Scheduling:
Optimal Unit Commitment: Constraints in unit commitment- Spinning Reserve, Thermal unit constraints and Hydro constraints. Cost function formulation: start-up cost

consideration and shutdown cost consideration. Priority ordering method for the solution of UC Problem. Optimal scheduling of Hydrothermal System -Short-term hydrothermal scheduling problem.

UNIT-III:

Modelling of Components: Basic Generator Control loops – ALFC and AVR loops and cross coupling between two loops. Modelling of Turbine: first order Turbine model, Block Diagram representation of Steam Turbines and Approximate Linear Models. Mathematical Modelling of Speed Governing System – Governor Characteristics, Regulation of two generators, Derivation of small signal transfer function and Generator load model.

UNIT-IV:

Load Frequency Control: Necessity of keeping frequency constant, concept of control area, Incremental power Balance of a control area, Block diagram representation of Single area system, Steady state analysis and Dynamic response for uncontrolled case. Proportional plus Integral control of single area and steady state and dynamic response. Modelling of a tie line, Load frequency control of two-area system – uncontrolled case and controlled case.

UNIT-V:

Reactive Power Control: Fundamental components of an Excitation system and its transfer function model, Overview of Reactive Power control, Reactive Power compensation in transmission systems Shunt and Series Compensation, Synchronous Condenser – advantages and disadvantages of different types of compensating equipment. Need for real time and computer control of power systems.

TEXT BOOKS:

1. Modern Power System Analysis, D. P. Kothari, I. J. Nagrath, 4th Edition, Tata McGraw-Hill Education Private Limited, 2011
2. Electric Energy Systems Theory, O. I. Elgerd, 2nd Edition, Tata McGraw-Hill Publishing Company Ltd.
3. Operation and Control in Power Systems, P. S. R. Murthy, 2nd Edition, BS Publications, 2011

REFERENCES:

1. Power System Generation, Operation and Control, Allen J. Wood, Bruce Wollenberg, Gerald B. Sheble, 3rd Edition, John Wiley and Sons, 2013
2. Power System Analysis, Grainger, Stevenson, Tata McGraw-Hill
3. Power System Operation and Control, S. Sivanagaraju, G. Sreenivasan, 5th Edition, Pearson
4. Power System Analysis, Hadi Saadat, TMH Edition

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/108/104/108104052/
2. https://nptel.ac.in/courses/108/105/108105133/
3. https://nptel.ac.in/courses/108/102/108102080/
4. https://nptel.ac.in/courses/108/106/108106026/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EE402) SWITCHED MODE POWER CONVERSION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Power Electronics, Electronic Devices and Circuits

COURSE OBJECTIVES:
- To understand various modes of operation of DC-DC converter
- To analyze control aspects of converter
- To design various Switched Mode Power Supply components
- To get awareness on EMI, protection of converter system

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand and analyze various modes of operation of DC-DC converter
CO-2: Design and model the components for DC-DC Converter along with EMI suppression
CO-3: Analyze various types of Resonant Converter

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	2	2	3	3	-	-	-	-	2	2	3	3
CO-2	3	3	3	3	3	2	2	-	-	-	2	2	3	3
CO-3	2	2	3	2	3	2	2	-	-	-	2	2	3	3

UNIT-I:
Introduction: Linear converters Vs switching converters. basic principles of switch mode power conversion, concept of steady state in switching converters, volt-second and ampere-second balance equations.
Non-Isolated Converters: Steady state analysis of Buck Converter, Boost Converter for both continuous and discontinuous conduction modes of operation.

UNIT-II:
Introduction to Isolated DC-DC converters: Steady state analysis of isolated Fly back Converter, Forward Converter, Half-Bridge and Full Bridge Converters, Continuous conduction mode converters, Choice of switching frequency.

UNIT-III:
Magnetic Design: Selection of energy storage inductor, Design of high frequency Inductor and high frequency transformer Design of Buck and Boost Converter.
Electro Magnetic Interference (EMI): EMI Filter Components, Conducted EMI suppression, Radiated EMI suppression, Measurement. Design aspects of EMI filters for switch mode power supplies

UNIT-IV:

Modelling of Converter: Basic converter circuit components modelling-Block diagram representation of Converter.

Modelling of system representation: State Space Model, Average Model of Converter and Transfer Function Model. State space model of RC, RLC Circuit and DC-DC Converter (Buck, Boost).

UNIT-V:

Resonant Converters: Introduction - Need of resonant converter- Classification of Resonant Converters-Basic Resonant Circuit Concepts- Load-resonant Converters **Resonant Transition Converters:** Zero Voltage Switching and Zero Current Switching converters.

TEXT BOOKS:

1. Fundamentals of Power Electronics, Robert W. Erickson, Dragan Maksimovic, Reprint of the Original 2nd Edition, Springer, 2012
2. Power-Switching Converters, Simon Ang, Alejandro Oliva, 3rd Edition, CRC Press, 2010
3. Power Electronics, M. H. Rashid, Prentice Hall of India

REFERENCES:

1. Power Electronics Converters, Application and Design, Mohan N., Undeland T., Robbins W., 3rd Edition, John Wiley, 2002
2. Design of Magnetic Components for Switched Mode Power Converters, Umanand L., Bhat S. R., Wiley Eastern, 1992
3. Lecture Note on Switched Mode Power Conversion, V. Ramanarayanan

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/108/108/108108036
2. https://www.analog.com/media/en/analog-dialogue/volume-54/number-2/speed-up-the-design-of-emi-filters-for-switch-mode-power-supplies.pdf

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EE403) ELECTRIC VEHICLE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Electrical Machines, Power Electronics

COURSE OBJECTIVES:
- To aware the evolvement of electric and hybrid electric vehicles
- To study the different drive train configurations of electric vehicles
- To propose the various propulsion and energy storage systems for EHVs
- To know the sizing of propulsion motors and other systems involved in EH vehicles

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Assess the performance, societal and environmental impact of EHVs having known their past history
CO-2: Implement various drive train topologies and control strategies in Electric and Hybrid vehicles
CO-3: Understand the Design/Sizing procedure and control of different electric propulsion units and other components of EHVs and EVs
CO-4: Select the energy storage system appropriately and strategize its management in EHVs and EVs

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	3	2	1	3	1	1	-	1	-	1	-	3	2
CO-2	2	3	2	1	3	1	1	-	1	-	1	-	3	2
CO-3	3	3	3	2	3	2	2	-	1	-	1	1	2	3
CO-4	3	2	2	1	2	-	1	-	-	-	-	-	3	2

UNIT-I:
Introduction to Electric Vehicles: History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drivetrains on energy supplies. Conventional Vehicles: Forces acting on a vehicle, vehicle power source (plant) characteristics, transmission characteristics: Manual, Hydro-dynamic and Continuously Variable Transmissions, mathematical model of vehicle, vehicle performance.

UNIT-II:
Electric Drive-Trains: Advantages of Electric Vehicles (EVs), Basic concept of electric vehicle traction, introduction to various electric drive-train topologies, Electric Vehicle Performance, power flow control, fuel efficiency analysis.

UNIT-III:

Hybrid Electric Drive-Trains: Basic architecture and concept of hybrid traction, patterns of power flow, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis, Concept of Drive Cycle.

UNIT-IV:

Electric Propulsion Unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives, drive system efficiency, axial flux machine topologies for EV applications.

UNIT-V:

Sizing of the Drive System: Matching the electric machine and the internal combustion engine (ICE), Sizing of the propulsion motor, Sizing of engine-generator, Sizing the power electronics based on Switch Technology - Switching Frequency and Ripple capacitor design, Supporting subsystems.

Energy Management Strategies-Introduction to energy management strategies in HEVs, classification and comparison of different energy management strategies.

TEXT BOOKS:

1. Electric and Hybrid Vehicles: Design Fundamentals, Iqbal Hussein, CRC Press, 2010
2. Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, CRC Press, 2009
3. Electric Vehicle Technology Explained, James Larminie, John Lowry, Wiley, 2003

REFERENCES:

1. Hybrid Vehicle Propulsion, Jefferson C. M., Barnard R. H., WIT Press, 2002
2. Hybrid, Electric and Fuel Cell Vehicles, Jack Erjavec and Jeff Arias, Cengage Learning, 2012
3. Electric Vehicles - The Benefits and Barriers, Seref Soylu, InTech Publishers, 2011
4. Alternative Fuel Technology – Electric, Hybrid and Fuel Cell Vehicles, Jack Erjavec and Jeff Arias, Cengage Learning, 2007
5. Build Your Own Electric Vehicle, Seth Leitman, McGraw-Hill, 2013

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EE422) SENSORS AND ACTUATORS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To expose to various sensors and transducers for measuring mechanical quantities
- To familiarize with the specifications of sensors and transducers
- To identifyfor various sensors and transducers for various applications
- To expose to various actuators

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Familiarize with classification and characteristics of various sensors
CO-2: Familiarize with the principle and working of various sensors
CO-3: Describe basic laws and phenomena that define behaviour of sensors and actuators
CO-4: Familiarize with the principle and working of various Actuators
CO-5: Select proper sensor/Actuator for a specific measurement application

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	2	2	2	2	2	2	2	2	2	1	2	2	2
CO-2	3	3	3	2	1	2	2	2	-	2	-	2	2	3
CO-3	2	2	2	2	2	2	3	1	2	-	-	3	3	2
CO-4	3	3	3	2	1	3	3	1	-	2	2	1	2	2

UNIT-I:
Sensors & Transducers: Principles, Classification, Parameters, Characteristics, Environmental Parameters (EP), Characterization. Mechanical and Electromechanical Sensors: Introduction, Resistive Potentiometer, Strain Gauge, Resistance Strain Gauge, Semiconductor Strain Gauges, Inductive Sensors- Sensitivity and Linearity of the Sensor, Types- Capacitive Sensors, Electrostatic Transducer, Force/Stress Sensors using Quartz Resonators, Ultrasonic Sensors.

UNIT-II:
Magnetic Sensors: Introduction, Sensors and the Principles Behind, Magneto-resistive Sensors, Anisotropic Magneto-resistive Sensing, Semiconductor Magneto-resistors, Hall Effect and Sensors, Inductance and Eddy Current Sensors, Angular/Rotary Movement Transducers, Synchros, Synchroresolvers, Eddy Current Sensors, Electromagnetic Flowmeter, Switching Magnetic Sensors, SQUID Sensors.

UNIT-III:
Radiation Sensors: Introduction – Basic Characteristics – Types of Photosensistors/Photo detectors– X-ray and Nuclear Radiation Sensors– Fiber Optic Sensors.

Electro Analytical Sensors: Introduction – The Electrochemical Cell – The Cell Potential –Standard Hydrogen Electrode (SHE) – Liquid Junction and Other Potentials.

UNIT-IV:

Smart Sensors: Introduction, Primary Sensors, Excitation, Amplification, Filters, Converters, Compensation, Information Coding/Processing, Data Communication, Standards for Smart Sensor Interface, the Automation. Sensors Applications: Introduction, On-board Automobile Sensors (Automotive Sensors), Home Appliance Sensors, Aerospace Sensors, Sensors for Manufacturing, Sensors for environmental Monitoring.

UNIT-V:

Actuators: Pneumatic and Hydraulic Actuation Systems- Actuation systems, Pneumatic and hydraulic systems, Directional Control valves, Pressure control valves, Cylinders, Servo and proportional control valves, Process control valves, Rotary actuators, Mechanical Actuation Systems Types of motion, Kinematic chains, Cams, Gears, Ratchet and pawl, Belt and chain drives, Bearings, Mechanical aspects of motor selection, Electrical Actuation Systems, Electrical systems, Mechanical switches, Solid-state switches, Solenoids, D.C. Motors, A.C. Motors, Stepper motors.

TEXT BOOKS:

1. Sensors and Transducers, D. Patranabis, PHI Learning Private Limited
2. Mechatronics, W. Bolton, Pearson Education Limited

REFERENCES:

1. Sensors and Actuators, Patranabis, 2nd Edition, PHI, 2013

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EC302) REAL TIME OPERATING SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Computer Organization and Design

COURSE OBJECTIVES:
- To learn the general embedded system concepts
- To understand design of embedded hardware and software development tools
- To learn the basics of OS and RTOS
- To describe key issues such as CPU scheduling, memory management, task synchronization, and file system in the context of real-time embedded systems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the hardware and software requirements of real time and non-real time embedded systems.
CO-2: Familiarize with the concepts of RTOS
CO-3: Analyze an embedded application with real time operating systems
CO-4: Understand scheduling and inter process communication concepts in real-time environment
CO-5: Interpret embedded systems design and develop process

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-2	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-3	2	3	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-4	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-5	2	3	3	-	-	-	-	-	-	-	-	-	-	-	3

UNIT-I:
Fundamentals of Embedded Systems: Definition - Characteristics and Design metrics of Embedded system, Classification of Embedded Systems, Real Time Systems - Need for Real-time systems, Hard and Soft Real-time systems, Processors in the system - Other Hardware units, Software components, Examples for embedded systems, Design issues and trends.

UNIT-II:
Embedded Software Development Environment: Embedded System Development Process, Programming languages, Software Development tools - Host and Target

machines, Linkers/Locators for embedded software, getting embedded software into the target system, Testing on host machine.

UNIT-III:

Real Time Operating Systems Concepts–I: Need for RTOS in embedded system, Comparison with GPOS, RTOS Architecture and Characteristics, Tasks and Task states, TCB, Context Switching, Task scheduling - Scheduling algorithms - Rate Monotonic, EDF, Round Robin, Round Robin with Interrupts, Priority driven – Preemptive and Non-preemptive scheduling.

UNIT-IV:

Real Time Operating Systems Concepts–II: Inter-Process Communication mechanisms - Semaphores - Message queues - Mailboxes -Pipes - Events - Timer functions, Task Synchronization - Shared data - Priority Inversion - Inheritance and Ceiling, Dead lock, Memory management, Interrupt routines in RTOS environment.

UNIT-V:

Design Examples and Case Studies: Case study of embedded system design and coding for Automatic Chocolate Vending machine using µCOS RTOS, Case study of Digital Camera Hardware and Software architecture.

TEXT BOOKS:

1. Embedded Systems Architecture, Programming and Design, Raj Kamal, 2nd Edition, Tata McGraw-Hill, 2011
2. An Embedded Software Primer, David E. Simon, 1st Edition, Pearson, 2005

REFERENCES:

1. Real-Time Systems, J. W. S. Liu, Pearson, 2009
2. Real-Time Embedded Systems: Design Principles and Engineering Practices, 1st Edition, Newnes, 2015
3. Computers as Components - Principles of Embedded Computing System Design, Wayne Wolf, 2nd Edition, Morgan Kaufmann, 2008

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2EE401) POWER SYSTEMS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Power Systems-I, Power Systems-II, Switchgear and Protection, Power System Analysis

COURSE OBJECTIVES:
- To perform simulation experiments in various software
- To observe the characteristics of IDMT, OV/UV, Differential relays
- To perform fault analysis on Generators, transformers, Transmission line models

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Perform load flow solution by using various methods
CO-2: Analyse different relays
CO-3: Test CT and PT's, insulator strings
CO-4: Determine sequence impedance and fault currents of Generator, transformer and transmission line models

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	3	3	2	1	1	3	3	3	2	3	3
CO-2	3	3	3	3	3	2	1	1	3	3	2	2	3	3
CO-3	3	3	3	3	3	2	1	1	3	3	2	2	3	3
CO-4	3	3	3	3	3	2	1	1	3	3	2	2	3	3

LIST OF EXPERIMENTS:
1. Characteristics of Electromagnetic IDMT over current relay.
2. Characteristics of Micro Processor based Over voltage/Under voltage relay.
3. Differential protection of 1-Φ transformer.
4. Testing of CT and PT's, insulator strings.
5. Fault location of underground cable.
6. Measurement of Capacitance of 3-core cables.
7. Determination of sequence impedances of a 3-Φ synchronous machine.
8. Determination of sequence impedances of a 3-Φ Transformer.
9. Fault analysis of transmission system.
10. Formation of Y BUS
11. Load flow analysis with GS Method.
12. Load flow analysis with FDLF method.
13. Load frequency control of two area system
14. Simulation of Renewable Energy based Electrical Power Generation using
    a) Solar Energy
    b) Wind Turbine

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2EC421) PRINCIPLES OF DIGITAL SIGNAL PROCESSING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: MATLAB on Ramp, Digital Signal Processing

COURSE OBJECTIVES:
- To simulate and implement simple programs on the DSP processor
- To verify properties of a discrete system
- To learn various transforms on digital signals
- To understand the design of digital filters
- To understand concepts to design the drives

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply knowledge of digital filter design for various applications
CO-2: Analyse various signals in transform domain
CO-3: Design digital filters using different transformation techniques
CO-4: Perform real time experiments on the processors

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	3	2	2	1	3	1	2	1	2	3	2
CO-2	3	3	3	1	1	1	1	3	1	2	1	2	3	2
CO-3	3	3	3	2	2	1	1	3	1	2	1	2	3	2
CO-4	3	3	2	3	3	3	3	3	2	2	1	2	3	2

LIST OF EXPERIMENTS:

The following experiments are to be performed using MATLAB:
1. Operations on signals and sequences such as Addition, Multiplication, Scaling, Shifting, Folding, Computation of Energy and Average Power.
2. Verification of Linearity and Time Invariance Properties of a given Continuous/ Discrete System.
3. Linear Convolution and Circular Convolution
4. Computation of Unit sample, Unit step and sinusoidal responses of the given LTI system and verifying its Physical realizability and stability properties.
5. Discrete Fourier Transform / Inverse Discrete Fourier Transform
6. Fast Fourier Transform
7. Sampling theorem Verification.
8. Implementation of Filters using IIR
9. Implementation of Filters using FIR

The following experiments are to be performed using DSP Processor Kit:

1. Generation of sine wave and square wave using DSP trainer kit
2. To Verify Linear Convolution and Circular Convolution
3. PWM generation on DSP training kit

TEXT BOOKS:

1. A Guide to MATLAB For Beginners and Experienced Users, Brian R. Hunt, Ronald L. Lipsman, Jonathan M. Rosenberg, 2001
2. Discrete Time Signal Processing, A. V. Oppenheim and R. W. Schaffer, PHI, 2009
3. TMS 320F 24xx Manuals

REFERENCES:

1. Digital Signal Processing - Fundamentals and Applications, Li Tan, Elsevier, 2008
2. Fundamentals of Digital Signal Processing using MATLAB, Robert J. Schilling, Sandra L. Harris, Thomson, 2007
3. Digital Signal Processing, S. Salivahanan, A. Vallavaraj, C. Gnanapriya, TMH, 2009
4. Discrete Systems and Digital Signal Processing with MATLAB, Taan S, El Ali, CRC Press, 2009
5. Digital Signal Processor: Architecture, Programming & Application, P. Venkata Ramani, M. Bhaskar, Tata McGraw-Hill, 2001

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE404) HVDC TRANSMISSION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To comprehend the conversion principles of HVDC transmission
- To analysis of 3, 6, 12 pulse converters, rectifier and inverter operations of HVDC converters
- To identify the different types of Harmonics and reduction by using filters
- To comprehend Interaction between HVAC and DC systems in various aspects
- To appreciate the reliable MTDC systems and protection of HVDC system

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand HVDC technology
CO-2: Apply the knowledge of modelling and analysis of HVDC system for inter-area power flow regulation
CO-3: Appreciate the reliable Multi terminal HVDC system
CO-4: Apply advanced protective schemes for HVDC systems against transient over voltages and over currents

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	2	1	3	3	3	-	2	1	-	2	2	3	3
CO-2	3	3	2	3	3	2	-	1	2	-	2	2	3	3
CO-3	3	3	3	2	3	3	-	1	2	-	2	2	3	3
CO-4	3	3	3	3	3	2	-	1	2	-	2	2	3	3

UNIT-I:
HVDC Technology: Historical development in DC Transmission, Advantages & Disadvantages of DC Transmission over Ac Transmission, DC Transmission Systems: Mono-polar, bi-polar and homo-polar lines, back-to-back HVDC systems, Components of HDVC Transmission System, Main applications of DC Transmission.

UNIT-II:
HVDC Converters: Rectifier and Inverter operation of converters, Line commutated Converters, Analysis of Graetz bridge converter with and without overlap, Analysis of 12 pulse converter. Characteristics and non-characteristics harmonics.

UNIT-III:
Converter and HVDC System Control: Converter control characteristics, System Control Hierarchy, Firing Angle Control, Current and Extinction Angle Control, Starting and stopping of DC Link, Power Control.

UNIT-IV:

MTDC Systems: Introduction-Potential applications of MTDC systems, Types of MTDC systems and its Comparison, Control and Protection of MTDC systems, Study of MTDC Systems.

UNIT-V:

Faults and Protection: Converter Faults, Protection against overcurrent's, Overvoltage's in converter station, Surge arresters, Protection against overvoltage's

Filters: Design of AC filters, Passive AC filters, DC Filters and Active Filters

TEXT BOOKS:

1. High Voltage Direct Transmission, J. Arrillaga, Peter Peregrinus,1983
2. HVDC Power Transmission Systems, K. R. Padiyar, Wiley Eastern,1990

REFERENCES:

1. Direct Current Transmission, E. W. Kimbark, Vol. I, Wiley Interscience,1971
2. Power Transmission, Direct Current, Erich Uhlmann, B. S. Publications, 2004
3. EHVAC and HVDC Transmission Engineering Practice, Theory, Practice and Solved Problems, Felix A. Farret, M. S. Rao, Khanna Publishers,1990

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE405) MICROGRID OPERATION AND CONTROL

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Power System, Power System Operation and Control

COURSE OBJECTIVES:
- To develop a comprehensive understanding of microgrid, its components, and operating principles
- To analyze the dynamic behavior of different microgrid components, including generation, storage, and load elements
- To investigate how droop control, voltage and frequency regulation, and power sharing algorithms are used to maintain stability and reliability in microgrid systems
- To investigate advanced control techniques for microgrid management

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand AC and DC microgrids fundamentals
CO-2: Analyze the Microgrid and carryout Dynamic modeling
CO-3: Distinguish different hierarchical control schemes for microgrid system and communication implementation
CO-4: Investigate the stability in microgrids and alternate options for improvement

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	3	3	3	2	-	1	-	1	1	2	3
CO-2	2	3	2	3	3	2	2	-	1	-	1	1	3	2
CO-3	3	3	2	2	3	3	2	-	1	-	1	1	2	2
CO-4	3	3	3	2	3	3	2	-	1	-	1	1	2	3

UNIT-I:
Introduction to Microgrid: Concept of microgrid, the overview of the structure and architecture of microgrid with brief control, operational aspects of DC and AC microgrids.

UNIT-II:
Modeling and Operation of Microgrid: concept of autonomous and non-autonomous microgrid, Dynamic modelling of individual components in AC and DC microgrids, state space modal analysis and influence of system parameters on the microgrid dynamics.

UNIT-III
Control Mechanism: DC microgrid control mechanism, droop control, issues in achieving active power sharing with impedance droop, remedies to achieve active power sharing, Transient frequency response, reactive power sharing and voltage regulation

UNIT-IV:
Multi-Microgrid Coordination and Control: review of sources of microgrid, hybrid microgrid: AC-AC, AC-DC and DC-DC microgrid clustering, coordinated control schemes in multi-microgrids

UNIT-V:
Control of Micro Grid System: Load Frequency Control (LFC) in Microgrid System – Voltage Control in Micro Grid System – Reactive Power Control in microgrid. Case Studies and Test beds for the microgrids

TEXT BOOKS:

1. Voltage-Sourced Converters in Power Systems: Modeling, Control, and Applications, Amirnaser Yazdani, Reza Iravani, John Wiley & Sons, 2010
2. Microgrids Architecture and control, IEEE Press Series, N. D. Hatziargyriou, 1st Edition, John Wiley & Sons, 2013
3. Microgrid Dynamics and Control, H. Bevrani, B. François, and T. Ise, 1st Edition, John Wiley & Sons, 2017

REFERENCES:

1. Cooperative Synchronization in Distributed Microgrid Control, Bidram, V. Nasirian, A. Davoudi, F. L. Lewis, 1st Edition, Springer, 2017
2. Power System Stability and Control, P. Kundur, 2nd Edition, McGraw-Hill, 2006

ONLINE RESOURCES:
1. DC Microgrid and Control System - Course

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE406) INTEGRATION AND CONTROL OF RENEWABLE ENERGY TECHNOLOGIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide necessary knowledge about the issues related to Integration of Renewable energy sources to the grid and Grid interfacing requirements
- To know the principles of static and dynamic energy conversion technologies
- To understand the control issues and challenges of various renewable systems Integration to grid

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the various issues related to Grid Integration and Interfacing
CO-2: Analyze the working of different renewable energy sources
CO-3: Discriminate the static and dynamic energy conversion technologies
CO-4: Apply the basic control strategies and issues required for grid integration

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	2	2	2	2	2	2	2	-	-	1	1	2	2
CO-2	3	3	3	2	1	2	2	2	-	-	1	1	2	3
CO-3	2	2	2	2	2	2	3	1	-	-	1	1	3	2
CO-4	3	3	3	2	1	3	3	1	-	-	1	1	2	2

UNIT-I:
Introduction: Electric grid introduction, power quality issues, Stability aspects, Grid Integration of Renewables, Challenges in Grid Integration, Balancing Mechanisms, Ancillary Services, Applications of Energy Storage Systems for Balancing Renewables. Importance and Effects of Renewable Energy penetration into the grid, Boundaries of the actual grid configuration.

UNIT-II:
Static Energy Conversion Technologies and Solar Integration: Introduction to different conventional and nonconventional static generation technologies, principle of operation and analysis of fuel cell, photovoltaic based generators, Operation and Control of Grid-Connected PV System, Control of Single-Phase PV System, Control of Three-Phase Grid-Connected PV system and integrated operation of different static energy conversion devices.

UNIT-III:
Dynamic Energy Conversion Technologies and Wind Integration: Introduction to different conventional and nonconventional dynamic generation technologies, principle of operation and analysis of reciprocating engines, gas and micro turbines,

hydro and wind-based generation technologies, Grid Integration of Wind Turbine System, Operation and Control of Grid-Connected Wind Energy System, and integrated operation of different dynamic energy conversion devices.

UNIT-IV:

Real and reactive power control: Control issues and challenges in Diesel, PV, wind and fuel cell-based generators, PLL, Modulation Techniques, filters for grid connected system, Linear and nonlinear controllers, Load frequency and Voltage Control.

UNIT-V:

Integration of Different Energy Conversion Technologies: Resources evaluation and needs, sizing of integration systems, Interfacing requirements, integrated Control of different resources, Distributed versus Centralized Control, Synchro Converters, Grid connected and Islanding Operations, Islanding detection methods.

TEXT BOOKS:

1. Operation and Control of Renewable Energy Systems, Mukhtar Ahmed, John Wiley & Sons, 2018
2. Integration and Control of Renewable Energy in Electric Power System, Ali Keyhani Mohammad Marwali and Min Dai, John Wiley & Sons, 2010
3. Power Conversion and Control of Wind Energy Systems, Bin Wu, Yongqiang Lang, Navid Zargari, Wiley 2011

REFERENCES:

1. Renewable and Efficient Electric Power Systems, G. Masters, IEEE-Wiley Publishers, 2013
2. Control of Power Inverters in Renewable Energy and Smart Grid Integration, Quing-Chang Zhong, Wiley, IEEE Press
3. Power Electronics: Converters, Applications, and Design, Ned Mohan, Tore M. Undeland, William P. Robbins, 3rd Edition, Wiley Publishers,1989

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE407) RELIABILITY ENGINEERING AND APPLICATIONS TO POWER SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Control Systems, Electrical Distribution Systems and Automation

COURSE OBJECTIVES:
- To describe Rules for combining probabilities of events and Binomial distribution
- To analyze Series, Parallel, Series-Parallel and Non-series parallel networks
- To describe Markov models and Frequency and duration concepts
- To apply Reliability concepts for Generation, composite and Distribution systems
- To evaluate basic and performance indices of radial networks

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Describe the reliability and unreliability rules, Reliability analysis of series parallel non series parallel system
CO-2: Illustrate the hazard rate function and expressions for different reliability functions
CO-3: Design the discrete Markov chains and continuous Markov process
CO-4: Apply Reliability concepts for Generation, composite and Distribution systems

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	1	-	-	2	-	-	-	2	1	2	3
CO-2	3	3	3	2	1	2	1	-	-	2	1	1	2	3
CO-3	-	2	3	1	2	2	1	2	2	3	2	1	2	2
CO-4	2	3	3	2	3	3	2	2	1	2	3	2	3	2

UNIT-I:
Basics of Probability Theory and Distribution: Concepts of Reliability, Unreliability, Availability, Unavailability–Rules for combining probabilities of events – Bernoulli's trials – probability density and distribution functions – Binomial distribution – expected value and standard deviation of binomial distribution

UNIT-II:
Network Modelling and Reliability Analysis: Analysis of Series, Parallel, Series-Parallel networks, complex networks: decomposition method, Path based and Cut-set based approaches, Problems.

UNIT-III:

Reliability Functions: Reliability functions f(t), F(t), R(t), h(t) and their relationships – Bathtub curve - Expected value and standard deviation of exponential distribution-reliability measures: MTTF, MTTR and MTBF.

UNIT-IV:

Markov Modelling Discrete Markov Chains: Concept of stochastic transitional probability Matrix, Evaluation of limiting state Probabilities, Continuous Markov process: one component repairable model – time dependent probability evaluation using Laplace transform approach – evaluation of limiting state probabilities using STPM

Frequency and Duration Techniques: Frequency and duration concept – Evaluation of frequency of encountering state-evaluation of cumulative probability and cumulative frequency of encountering of merged states of two component repairable model.

UNIT-V:

Generation System Reliability Analysis: Reliability model of a generation system: Recursive relation for unit addition and removal methods, load modelling - Merging of generation and load model Probability and frequency of failure for merged state model, Problems

Composite System Reliability Analysis: Markov model- Weighted average rate model– Decomposition method – Reliability Indices. Distribution System Reliability Analysis: Evaluation of Basic and performance indices of radial networks, Problems.

TEXT BOOKS:

1. Reliability Evaluation of Engineering System, R. Billinton, R. N. Allan, BS Publications
2. Reliability Evaluation of Power Systems, R. Billinton, R. N. Allan, BS Publications

REFERENCES:

1. Reliability Engineering: Theory and Practice, Alessandro Birolini, Springer Publications
2. An Introduction to Reliability and Maintainability Engineering, Charles Ebeling, TMH Publications
3. Reliability Engineering, E. Balaguruswamy, TMH Publications
4. Reliability Engineering, Elsayed A. Elsayed, Prentice Hall Publications
5. System Reliability Concepts, V. Sankar, Himalaya Publications

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC420) CPLD AND FPGA ARCHITECTURES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Switching Theory and Logic Design

COURSE OBJECTIVES:
- To introduce digital design concepts through various Programmable Logic Devices
- To understand the CPLD and FPGA architectures in detail
- To analyze the physical design cycle in FPGA
- To know the various applications of CPLD and FPGAs

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Design digital applications using PLDs
CO-2: Analyze the architectural features of CPLDs, FPGAs
CO-3: Analyze Physical Design cycle for FPGA
CO-4: Implementation of various applications using FPGA

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)***

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	-	-	-	-	-	-	-	-	-	2	3
CO-2	3	2	3	-	-	-	-	-	-	-	-	-	2	3
CO-3	3	3	3	-	-	-	-	-	-	-	-	-	2	3
CO-4	3	2	3	-	-	-	-	-	-	-	-	-	2	3

UNIT-I:
Introduction to Programmable Logic Devices: Programmable logic devices (PLD), Simple Programmable Logic Devices (SPLD) – Read Only Memories, Programmable Logic Arrays (PLA), Programmable Array Logic (PAL): Registered PALs, Configurable PALS, Digital design using PLDs.

UNIT-II:
Complex Programmable Logic Devices: Features and applications of complex programmable logic devices, Altera Max - 7000 series and Altera FLEX logic- 10K series CPLD, Xilinx Cool Runner - II CPLD, CPLD Implementation of a parallel adder with accumulation.

UNIT-III:

Field Programmable Gate Arrays: Features and applications of FPGAs, advantages and disadvantages of FPGA, architecture of FPGA, technology trends, programming technologies, commercially Available FPGAs.

UNIT-IV:

SRAM Field Programmable Gate Arrays: SRAM Programming Technology, SRAM Programmable FPGAs: Xilinx XC4000, Spartan-3 FPGA Architectures.
Anti-Fuse Programmed FPGAs: Anti-fuse Programming technology, The Actel ACT1, ACT2 and ACT3 architectures.

UNIT-V:

Physical Design Implementation on FPGAs: FPGA Design flow, Physical Design cycle for FPGAs, Partitioning, Routing-non-segmented, segmented and staggered models.
Design Applications: General design issues, Counter design Adder design using FPGA.

TEXT BOOKS:

1. Fundamentals of Logic Design, Charles H. Roth Jr, 5th Edition, Cengage Learning, 2004
2. Field Programmable Gate Array Technology, Stephen M. Trimberger, Springer International Edition, 1994

REFERENCES:

1. Algorithms for VLSI Physical Design Automation, Naveed Sherwani, 3rd Edition, Springer International Edition, 2005
2. Field-Programmable Gate Arrays, Stephen D. Brown, Springer, 1992

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE408) POWER QUALITY AND FACTS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Power Systems, Power Electronics and Electrical Drives

COURSE OBJECTIVES:
- To get the knowledge on power quality issues
- To get the knowledge on PQ mitigation methods
- To understand the need of various FACTS controllers
- To understand the working of combined compensators

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Acquire knowledge of power quality issues and fundamentals of FACTS devices
CO-2: Understand the Power Quality important methods
CO-3: Analyze the working and control of Shunt FACTS devices
CO-4: Study the working and control of series and combined compensators

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	2	1	1	2	1	-	1	-	-	1	2	2	2
CO-2	3	2	1	2	3	1	-	1	-	-	1	2	2	3
CO-3	3	2	1	2	2	1	-	1	-	-	1	2	2	3
CO-4	3	2	1	2	2	1	-	1	-	-	1	2	2	3

UNIT-I:
Introduction to Power Quality
Power Quality Problems in Distribution Systems: Transient and Steady state variations in voltage and frequency, Unbalance, Voltage - Sags, Swells, Interruptions
Wave-form Distortions: harmonics, noise, notching, dc-offsets, fluctuations, Flicker and its measurement. Power Quality monitoring standards.

UNIT-II: Harmonics and Mitigation Methods
Harmonics: Effect of harmonics-harmonic distortion-voltage and current distortion-harmonic indices-inter harmonics. Harmonic sources from commercial and industrial loads, locating harmonic sources. Harmonics Vs transients.
Mitigation Methods: Overview of mitigation methods–from fault to trip, reducing the number of faults, reducing the fault clearing time, different events and mitigation methods.

UNIT-III:
Introduction to FACTS Controllers: Basics of Power Flow in AC transmission, Analysis of uncompensated AC transmission lines. Classification of FACTS Controllers based on

type of connection, relative importance of different types of controllers and benefits of FACTS Technology.

UNIT-IV:

Static Shunt Compensators: Objectives of Shunt compensation, Methods of controllable VAR generation, Static Var compensator, its characteristics, TCR, TSC, FC-TCR, TSC - TCR configurations, STATCOM, basic operating principle, control approaches and characteristics.

UNIT-V:

Static Series and Combined Compensators: Objectives of series compensators, variable impedance type of series compensators, TCSC, TSSC-operating principles and control schemes, working of SSSC, UPFC and IPFC

TEXT BOOKS:

1. Electrical Power System Quality, Roger C. Durgan, McGrahan Santoso & Beaty, McGraw-Hill, 2012
2. Understanding Power Quality Problems, Math H. J. Bollen, IEEE Press
3. Understanding FACTS: Concepts and Technology of Flexible AC Transmission Systems, Narain G. Hingorani & Laszlo Gyugyi, Wiley

REFERENCES:

1. Power Quality: Problems and Mitigation Techniques, Bhim Singh, Ambrish Chandra, Kamal Al-Haddad, John Wiley & Sons, 2015
2. FACTS Controllers in Power Transmission and Distribution, K. R. Padiyar, New Age International, 2007

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc23_ee58/preview
2. https://onlinecourses.nptel.ac.in/noc21_ee103/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE409) ENERGY AUDITING AND CONSERVATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To have acquaintance with Energy Conservation Act and to understand energy management principles
- To use energy efficient technologies like energy efficient motors and transformers
- To understand the methods of energy audits
- To know the operation of energy audit equipment

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the current energy scenario and importance of energy conservation
CO-2: Implement different energy management strategies
CO-3: Apply various electrical energy efficient technologies
CO-4: Carry out energy audits in industrial and lighting sectors

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	1	2	2	2	2	2	2	1	1	-	-	1	2	2
CO-2	1	3	2	2	2	2	2	1	1	-	1	1	2	2
CO-3	1	2	2	2	2	2	2	1	1	-	-	1	2	2
CO-4	1	3	2	3	2	2	2	1	1	-	3	1	2	2

UNIT-I:
Energy Scenario and Energy Management: Energy Conservation and its importance, energy conservation schemes long term energy scenario, energy pricing, reforms in energy sector, energy security, salient features of Energy Conservation Act-2001. Principles of Energy management, Qualities and Functions of energy manager, checklist for top management, questionnaire

UNIT-II:
Principles of Energy Audit: Energy audit- concept and definitions, types of audits, energy index, cost index, pie charts, Sankey diagrams, load profiles, Energy conservation schemes- Energy audit of thermal power station and substation as case studies

UNIT-III:
Energy Economics: Electricity billing/tariff, electrical Load management, Maximum Demand Control, Power factor improvement, its benefits, selection and location of capacitors, Automatic Power Factor controllers. Demand side management.

UNIT-IV:

Electrical Energy Efficient Technologies: Energy efficient motors, factors affecting efficiency, characteristics - variable speed, variable duty cycle systems, RMS hp-voltage unbalance- motor energy audit

Soft starters with energy saver-comparison with conventional starters, variable speed drives, energy efficient transformers, use of amorphous core, electronic ballast, use of LEDs.

UNIT-V:

Energy Audit in Illumination and Energy Audit Instruments: Design and practice of good lighting system, energy efficient lighting control, light energy audit.

Energy audit instruments: wattmeter, data loggers, thermocouples, pyrometers, flux meters, tongue testers.

TEXT BOOKS:

1. Energy Management, W. R. Murphy & G. Mckay, Butterworth-Heinemann
2. Energy Management, Paul O' Callaghan, 1st Edition, McGraw-Hill,1998

REFERENCES:

1. Energy Efficient Electric Motors, John C. Andreas, 2nd Edition, Marcel Dekker, 1995
2. Energy Management Handbook, W. C. Turner, John Wiley and Sons
3. Energy Management and Good Lighting Practice: Fuel Efficiency, Booklet, EEO
4. Guidebooks for National Certification Examination for Energy Manager/Energy Auditor books 1 and 3
5. Utilization of Electrical Energy and Conservation, S. C. Tripathy, McGraw-Hill 1991

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1AE306) FUEL CELL TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Engineering Chemistry, Fundamentals of Electrical and Electronics and Automotive Engines

COURSE OBJECTIVES:
- To understand the concepts, principle and working of fuel cell
- To study fuel cell operational calculations
- To present the components and automotive applications of fuel cell
- To outline the fueling techniques for fuel cell
- To provide the performance and analysis of fuel cell

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain the concepts, principle and working of fuel cell
CO-2: Compute the fuel processing calculations and cost estimation
CO-3: Describe the components and automotive applications of fuel cell
CO-4: Discuss the fueling techniques for fuel cell
CO-5: Analyze the performance of fuel cell

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	-	-	2	2	2	-	-	2	-	-	3	3
CO-2	3	3	3	2	2	2	2	-	-	2	-	-	3	3
CO-3	3	3	-	-	2	2	2	-	-	2	-	-	3	3
CO-4	3	3	-	-	2	2	2	-	-	2	-	-	3	3
CO-5	3	3	-	-	2	2	2	-	-	2	-	-	3	3

UNIT-I:
Introduction to Fuel Cells: Introduction, working and types of fuel cell, low, medium and high temperature fuel cell, liquid and methanol types, proton exchange membrane fuel cell solid oxide, hydrogen fuel cells, thermodynamics and electrochemical kinetics of fuel cells.

UNIT-II:
Sample Calculations: Unit operations - fuel cell calculations, fuel processing calculations, power conditioners; System issues - efficiency calculations,

thermodynamic considerations; cost calculations - cost of electricity, capital cost development, common conversion factors and automotive design calculations.

UNIT-III:

Fuel Cell Components and their Impact on Performance: Fuel cell performance characteristics, current/voltage, voltage efficiency and power density, Ohmic resistance, kinetic performance, mass transfer effects, membrane electrode assembly components, fuel cell stack, bi-polar plate, humidifiers and cooling plates.

UNIT-IV:

Fueling: Hydrogen storage technology – pressure cylinders, liquid hydrogen and metal hydrides, methods of hydrogen production, green hydrogen, carbon fibers; reformer technology – steam reforming, partial oxidation, auto thermal reforming, CO removal and fuel cell technology based on removal like biomass.

UNIT-V:

Fuel Cells for Automotive Applications: Fuel cells for automotive applications, technology advances in fuel cell vehicle systems, onboard hydrogen storage, liquid hydrogen and compressed hydrogen, metal hydrides, fuel cell control system, alkaline fuel cell and road map to market.

Solar Vehicle: Solar photovoltaic cell, solar array, solar car electrical system and drivetrain.

TEXT BOOKS:

1. Fuel Cells: Principles and Applications, Viswanathan B. and Scibioh Aulice M., University Press, 2006
2. Fuel Cells for Automotive Applications – Professional Engineering Publishing, 2004
3. Fuel Cell Handbook, 7th Edition, EG&G Technical Services, 2016

REFERENCES:

1. PEM Fuel Cells: Theory and Practice, Frano Barbir, Elsevier Academic Press, 2005
2. Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay and Ali Emadi, CRS Press, 2004
3. Fuel Cell Technology Handbook SAE International, Gregor Hoogers, 1st Edition, CRC Press, 2003

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/103/102/103102015/
2. https://onlinecourses.nptel.ac.in/noc23_ge47/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE410) HIGH VOLTAGE ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PREREQUISITES: Electro Magnetic Field Theory, Circuit Theory, Advanced Physics

COURSE OBJECTIVES:
- To understand the Gaseous, liquid and solid dielectric behavior under High Voltage
- To understand the generation methods of High A.C, D.C & Impulse Voltages required for various applications
- To understand the measuring techniques of High A.C., D.C & Impulse voltages and currents
- To understand the testing techniques for High Voltage Equipment

COURSE OUTCOMES: After completion of this course the student is able to
CO-1: Understand various applications and breakdown phenomenon in dielectric medium
CO-2: Realize the necessity to generate High A.C, D.C & Impulse voltages
CO-3: Analyze the measuring and testing techniques for High A.C, D.C and Impulse voltages
CO-4: Appreciate overvoltage phenomenon and basics of insulation coordination.

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	1	3	3	-	3	2	2	-	-	-	-	1	2	2
CO-2	1	3	1	2	3	3	3	1	-	-	2	1	2	2
CO-3	1	2	3	2	3	2	2	2	-	-		1	2	2
CO-4	1	3	3	2	2	1	1	1	-	-	1	1	2	2

UNIT-I:
Introduction to High Voltage Technology and Applications: Electric Field Stresses, Gas / Vacuum as Insulator, Liquid Dielectrics, Solids and Composites, Estimation and Control of Electric Stress, Numerical methods for electric field computation, Surge voltages, their distribution and control, Applications of insulating materials in Transformers, Rotating machines, Circuit Breakers, Cables and Bushings.

UNIT-II:
Break Down in Gaseous, Liquid and Solid Dielectrics: Gases as Insulating media, Collision process, Ionization process, Townsend's criteria of breakdown in gases, Paschen's law. Liquid as Insulator, Breakdown in pure and commercial liquids. Intrinsic

breakdown, Electromechanical breakdown, Thermal breakdown, Breakdown of solid dielectrics in practice, Breakdown in composite dielectrics.

UNIT-III:

Generation of High Voltages and Currents: Generation of High Direct Current Voltages, Generation of High Alternating Voltages, Generation of Impulse Voltages, Generation of Impulse currents, Tripping and Control of Impulse Generators.

UNIT-IV:

Measurement of High Voltages and Currents: Measurement of High Direct Current Voltages, Measurement of High Voltages Alternating and Impulse, Measurement of High Currents - Direct, Alternating and Impulse, Oscilloscope for Impulse Voltage and Current Measurements.

UNIT-V:

Non – Distructive Testing of Material and Electric Apparatus, Over Voltage Phenomenon and Insulation Coordination: Measurement of D.C Resistivity, Measurement of Dielectric Constant and loss factor, Partial discharge measurements. Testing of Isolators and Circuit Breakers, Testing of Cables, Testing of Transformers, Testing of Surge Arresters, Radio Interference Measurements, High voltage laboratory layout.

TEXT BOOKS:

1. High Voltage Engineering, M. S. Naidu and V. Kamaraju, 3rd Edition, TMH Publications
2. High Voltage Engineering: Fundamentals, E. Kuffel, W. S. Zaengl, J. Kuffel, 2nd Edition, Elsevier

REFERENCES:

1. High Voltage Engineering, C. L. Wadhwa, New Age Internationals, 1997
2. High Voltage Insulation Engineering, Ravindra Arora, Wolfgang Mosch, New Age International, 1995
3. Extra High Voltage A. C. Transmission Engineering, Rakosh Das Begamudre, New Age International, Revised Edition, 2007

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EE411) SMART GRID

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Basics of Power System

COURSE OBJECTIVES:

- To provide a foundational understanding of the Smart Grid concept in the context of modern energy systems
- To explore the advanced technologies that enable the functionality, efficiency, and intelligence of the Smart Grid
- To provide an understanding of Phasor Measurement Units (PMUs) and their role in modern power systems
- To address the smart grid security concerns through reliable communication technologies

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Appreciate the difference between smart grid & conventional grid
CO-2: Apply smart metering concepts to industrial and commercial installations
CO-3: Formulate solutions in the areas of smart substations, distributed generation and wide area measurements
CO-4: Understand Power Quality issues in Smart Grids and propose the solutions using modern communication technologies

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	3	3	3	3	3	2	1	-	-	2	1	3	3
CO-2	2	3	3	3	3	3	2	1	-	-	2	1	3	2
CO-3	2	3	3	3	3	3	2	1	-	-	2	1	2	3
CO-4	2	3	3	3	3	3	2	1	-	-	2	1	3	3

UNIT-I:

Introduction to Smart Grid: Introduction, Need of Smart Grid, Functions of Smart Grid, Opportunities of Smart Grid, Challenges in Implementation of Smart Grid, Difference between Conventional and Smart Grid, Concept of Resilience.

UNIT-II:

Smart Grid Technologies: Introduction to Smart Meters, Smart Appliances, Automatic Meter Reading (AMR), Advanced Metering Infrastructure (AMI), Outage Management System (OMS), Smart Sensors, Smart Substations, Substation Automation, Feeder Automation.

UNIT-III:
Phasor Measurement Unit (PMU): Features of Phasor measurement unit, Fundamentals, Global Positioning Satellite (GPS) Systems, Synchro phasor–Definition, Measurements, Applications of PMUs in Power Systems.
Comparisons between SCADA system and PMUs System, Intelligent Electronic Devices (IED)-Functions and Advantages.

UNIT-IV:
Power Quality in Smart Grid: Power Quality in Smart Grid, Causes of Poor Power Quality, Power Quality issues of Grid connected Renewable Energy Sources, Web based Power Quality monitoring, Concept of Load frequency control(LFC) and Voltage control in Microgrid system.

UNIT-V:
Communication Technologies in Smart Grid: Introduction to Communication Technology, Two Way Digital Communications Paradigm, Local Area Network (LAN)-Objectives, Home Area Network (HAN), Wide Area Network (WAN), Broadband Over Power Lines (BPL)-Features, Cyber Security for Smart Grid, Introduction to Internet of things (IoT)- Applications of IoT in Smart Grid.

TEXT BOOKS:

1. Smart Grid: Fundamentals of Design and Analysis, James Momoh, 1st Edition
2. Introduction to the Smart Grid Concepts, Technologies and Evolution, S. K. Salman, IET library, 2017
3. Design of Smart Power Grid Renewable Energy Systems, Ali Keyhani, Wiley IEEE, 2011

REFERENCES:

1. Synchronized Phasor Measurements and their Applications, A. G. Phadke and J. S. Thorp, Springer Edition, 2010
2. Gil Masters, Renewable and Efficient Electric Power System, Wiley-IEEE Press, 2004
3. Smart Grid: Technology and Applications, Janaka Ekanayake, Nick Jenkins, Kithsiri Liyanage, Wiley, 2012

ONLINE RESOURCES:
1. https://nptel.ac.in/courses/108107113
2. https://nptel.ac.in/courses/117107148